NATIONAL EXIT EXAM

ሰኔ (2017) JUNE (2025)

Computer Science

Questions (21-32)

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21. **Which one of the following is true about the percept of an agent?**

A. It refers to information generated by the agent program about possible actions to be taken by the agent.

B. It is an information captured from the environment by the agent sensor during execution phase.

C. It refers to seed knowledge provided to the agent during design phase.

D. It refers to the internal state memory that models how the agent world is evolved so far.

The correct answer is B. It is an information captured from the environment by the agent sensor during execution phase.

In Artificial Intelligence, a percept refers to the specific sensory input an agent receives at any given instant. While the agent operates, its sensors (like cameras, microphones, or data parsers) "perceive" the state of the environment, and this raw data is the percept.

22. **Which one of the following must satisfied to deploy a homogeneous distributed database?**

A. All local DBMS and users may use different software.

B. All device in which the local DBMS should have the same type of network architecture.

C. All local DBMS should be similar and users must use similar type of software.

D. Local DBMS may use different software but all users use identical software.

The correct answer is C. All local DBMS should be similar and users must use similar type of software.

In a homogeneous system, every site in the network uses the same DBMS software (e.g., all nodes are running Oracle 19c or all are running PostgreSQL 15). To the user, the entire system looks and acts like a single, unified database because there are no compatibility hurdles between the different locations.

23. **Which one of the following is correctly describes a distributed database?**

A. Data and software are distributed over multiple sites connected by some form of communication network.

B. It is a database system that put the data in central server and share its service through communication network.

C. It is a collection of Internet Web files connected through hyperlink.

D. It is a collection of logically inter-related tables resides in a single device.

The correct answer is A. Data and software are distributed over multiple sites connected by some form of communication network.

A Distributed Database (DDB) isn't just one big file sitting on a server; it is a collection of multiple, logically interrelated databases located at different physical sites. These sites are linked by a network (like a LAN or WAN) and managed by a Distributed Database Management System (DDBMS).

24. **Which one of the following describes a transition diagram of finite automaton?**

A. Directed graph associated with an FA.

B. Rows correspond to states.

C. Entries correspond to next states.

D. Columns correspond to inputs.

The correct answer is A. Directed graph associated with an FA.

A transition diagram is the visual, "map-like" representation of a Finite Automaton. In graph theory terms, it is a directed graph because:

- **Nodes (Vertices): Represent the states of the automaton (usually drawn as circles).**
- **Directed Edges (Arcs): Represent the transitions from one state to another (drawn as arrows).**
- **Labels: The arrows are labeled with the input symbols that trigger the transition.**

25. **Which one of the following software processes model uses both linear and sequential approach?**

A. Waterfall model

B. Spiral model

C. Agile model

D. V-Model

The correct answer remains A. Waterfall model.

In Software Engineering, the Waterfall model is officially known as the "Linear Sequential Model" (or the Classic Life Cycle).

- **Linear: It progresses in a straight line from start to finish.**
- **Sequential: Each phase must be completed in a specific order (sequence) before the next phase can begin.**

26. **Which one of the following describes the difference between multiprogramming batch system and time sharing system?**

A. Multiprogramming is efficient for many users to share a large computer than time sharing.

B. The objective of multiprogrammed batch systems is minimizing response time whereas in time-sharing systems is to maximize processor use.

C. The time sharing system shared the CPU between several processes from a separate terminal whereas multiprogramming read several jobs in

the single memory.

D. Multiprogramming allocate fixed time interval whereas multiprogramming applied CPU scheduling.

The correct answer is C. The time sharing system shared the CPU between several processes from a separate terminal whereas multiprogramming read several jobs in the single memory.

- **Time-Sharing Systems: These are designed for user interaction. The CPU switches between multiple users (often at separate terminals) so frequently that each user feels like they have the entire machine to themselves. The focus is on minimizing response time.**
- **Multiprogramming Batch Systems: These are designed for efficiency and throughput. Several "jobs" (tasks) are loaded into memory at once. If the current job has to wait (for example, for an I/O operation), the CPU simply switches to another job in memory to ensure the processor isn't sitting idle. The focus is on maximizing CPU utilization.**

27. **Which one of the following rules define a correctly structured JavaScript program?**

A. JavaScript analyzer

B. JavaScript Syntax

C. JavaScript Operators

D. JavaScript Semantics

The correct answer is B. JavaScript Syntax.

In programming, Syntax is the set of rules that defines the combinations of symbols and characters that are considered to be a correctly structured program. If the syntax is wrong, the code will fail to even start running.

28. **Which one of the following task is used to create the part of the website that users directly interact within their web browsers?**

A. Backend Development

B. Server Communication

C. Client-side programming

D. Server-side programming

The correct answer is C. Client-side programming.

In web development, the "Client" refers to the user's web browser (like Chrome, Safari, or Firefox). Client-side programming involves writing code, primarily HTML, CSS, and JavaScript, that is sent to the browser and executed there. This is what creates the visual interface, buttons, forms, and animations that users see and touch.

29. **Which one of the following is not an element of finite automaton?**

A. Set of input alphabet

B. Transition function

C. Set of states

D. Context-Free Grammar

The correct answer is D. Context-Free Grammar.

A Finite Automaton (FA) is formally defined as a 5-tuple. A Context-Free Grammar (CFG) is a completely different level of the Chomsky Hierarchy and is not an internal component of a Finite Automaton.

To be a complete Finite Automaton, the following five elements must exist:

1. Set of states: A finite set of all possible states the machine.
2. Set of input alphabet: A finite set of symbols (like {0,a},{1,b}) that the machine can read.
3. Transition function: The rulebook that tells the machine which state to move to based on the current state and the input symbol.
4. Start state: The specific state where the machine begins its operation.
5. Set of Accept/Final states: The states that, if reached at the end of the input, mean the string is accepted.

30. Which one of the following user privilege controls all services and allows user to make changes and check the activities of other users?

A. Regular

B. Guest

C. Child

D. Administrator

The correct answer is D. Administrator.

In almost all operating systems and network environments, the Administrator (sometimes called "Root" or "Superuser") holds the highest level of permissions. This role is designed to manage the entire system rather than just perform daily tasks.

31. **Which one of the following uninformed search algorithm is memory efficient comparing to others?**

A. Breadth first search

B. Depth first search

C. Uniform cost search

D. Bidirectional search

The correct answer is B. Depth first search.

Among the listed uninformed search algorithms, Depth First Search (DFS) is the most memory-efficient because it only needs to store a single path from the root to a leaf node, along with the remaining unexpanded sibling nodes for each node on the path.

32. **Which one of the following type of cache has relatively high speed than the others?**

A. L1 cache

B. L2 cache

C. L3 cache

D. Virtual cache

The correct answer is A. L1 cache.

In the memory hierarchy of a computer, the closer the memory is to the CPU cores, the faster it operates. L1 (Level 1) cache is the fastest and smallest memory bank, usually built directly into the processor chip itself.